SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Field	Details
Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	J. Thushany
Student ID	JF24ICT09
Implementation Status	In Progress
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators need a convenient way to export student records from the Student Management System for reporting, record keeping, and use in spreadsheet applications. Manually copying student information from the dashboard is time-consuming and may introduce errors.

The change request aims to provide a CSV export facility within the administrator dashboard.

1.2 Existing Behaviour

Before implementation:

- The administrator dashboard retrieved student records through the existing GET /api/students/ endpoint.
- Student information was displayed in a table.
- Administrators could search for, add, edit, and delete student records.
- No dedicated CSV export facility was available.
- Administrators had to rely on the dashboard or manually copy student information when preparing external records.

1.3 Requested Change

A Student Records Export to CSV feature is required in the administrator dashboard.

The feature must:

- Provide an **Export CSV** button for administrators.
- Export Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
- Generate a downloadable file named students.csv.
- Use clear column headers.
- Export all loaded student records, not only records currently visible after applying a search filter.
- Handle commas, quotation marks, and other special characters correctly in CSV fields.
- Handle an empty student list without causing an application error.
- Preserve the existing student search, Add, Edit, and Delete functionality.
- Remain available only through the existing administrator-protected student dashboard.

2. Implementation

2.1 Implementation Summary

The CR-015 enhancement adds a CSV export facility to the administrator dashboard. The administrator can download student records as a CSV file and open the file in spreadsheet applications that support CSV format.

The implementation uses the student records already loaded by the existing frontend. Therefore, a separate backend export endpoint is not required for the planned client-side export implementation.

The existing student API already returns the fields required for the CSV export. The API route also uses authentication and administrator-role authorization.

2.2 Affected Files

File	Change	Purpose
frontend/src/pages/DashboardPage.jsx	Modified	Add CSV generation, the Export CSV button, file download handling, and export feedback.
frontend/src/styles/dashboard.css	Modified	Add styling for the Export CSV button and its layout.

Additional files: None planned.

2.3 Technical Changes

CSV Column Structure

The exported file will contain the following headers:

1. Student ID
2. Registration Number
3. First Name
4. Last Name
5. Email
6. Phone Number

CSV Formatting

The implementation will format student values as CSV fields. Fields will be escaped appropriately so that commas and quotation marks within student information do not incorrectly split columns or records.

A UTF-8 byte-order mark (BOM) may be included to improve compatibility when opening the file in spreadsheet applications.

File Download

The export functionality will generate a CSV Blob in the browser and initiate a download using the filename:

students.csv

Export Scope

The export will use the complete student list already loaded into the dashboard, rather than the search-filtered list. This ensures that applying a search does not unintentionally restrict the exported records.

Empty-List Handling

When no student records are available, the export should still produce a valid CSV file containing the column headers, without causing an application error.

Existing CRUD Preservation

The existing Add, Edit, and Delete operations will remain unchanged. Student records will continue to use student_id as the identifier for record-specific operations.

2.4 Implementation Logic

1. Retrieve student records using the existing GET /api/students/ request.
2. Store the retrieved student records in the existing dashboard state.
3. Display the Export CSV button in the administrator dashboard.
4. When the administrator clicks the button, select the six required student fields.
5. Generate the CSV header row and corresponding student data rows.
6. Escape field values to preserve valid CSV formatting.
7. Create a downloadable CSV file named students.csv.
8. Initiate the browser download.
9. Display suitable feedback for successful export or an empty student list.
10. Preserve existing search, Add, Edit, and Delete functionality.

2.5 Error and Data Handling

The export implementation will use the existing student data-loading and error-handling mechanisms.

The following situations will be considered:

- Student records are available.
- No student records are available.
- Student information contains commas or quotation marks.
- The student list is still loading or has failed to load.
- The administrator attempts to export while the export button is disabled.

The export operation will not modify student records or update the database.

3. Testing & Verification

3.1 Testing Approach

Testing will be conducted after all CR-015 code changes are completed.

Testing Types:

- Functional testing
- Positive testing
- CSV format and data validation
- Empty-data testing
- UI testing
- Regression testing
- Existing CRUD verification
- Administrator authorization verification

Environment: Development

Browser: Google Chrome

Test Role: Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Verify Export CSV button	Button appears in the administrator dashboard.	Pending	Pending
TC-02	Export student records	A file named students.csv is downloaded.	Pending	Pending
TC-03	Verify CSV column headers	All six required headers appear in the correct order.	Pending	Pending
TC-04	Verify exported student data	Exported values match the corresponding student records.	Pending	Pending
TC-05	Export while a search filter is active	All loaded student records are exported.	Pending	Pending
TC-06	Export an empty student list	A valid CSV file with headers is downloaded without an application error.	Pending	Pending

TC-07	Verify special characters	Commas and quotation marks are correctly escaped.	Pending	Pending
TC-08	Verify existing Add, Edit, and Delete operations	Existing CRUD functionality continues to work correctly.	Pending	Pending
TC-09	Verify administrator authorization	The student-list API remains protected by authentication and administrator-role authorization.	Pending	Pending

3.3 Test Summary

Test Result	**Count**
Total Test Cases	9
Passed	Pending
Failed	Pending
Not Yet Tested	9
Pass Rate	To be calculated after testing

Overall Test Result: Pending

4. Acceptance Criteria Verification

ID	**Acceptance Criterion**	**Verification Method**	**Status**
AC-001	Administrators can initiate CSV export from the dashboard.	UI testing	Pending
AC-002	The exported file is named students.csv.	Download verification	Pending
AC-003	All six required columns are included.	Header inspection	Pending
AC-004	Exported records contain the correct student data.	Data comparison	Pending
AC-005	All loaded records are exported regardless of the active search filter.	Filtered-export testing	Pending

AC-006	Empty student data is handled without an application error.	Empty-data testing	Pending
AC-007	CSV special characters are escaped correctly.	CSV format testing	Pending
AC-008	Existing CRUD operations remain functional.	Regression testing	Pending
AC-009	Administrator authorization remains enforced.	Authorization verification	Pending

5. Evidence

Evidence 01 – CSV Export Functionality

Related: AC-001, AC-002, AC-003, AC-004

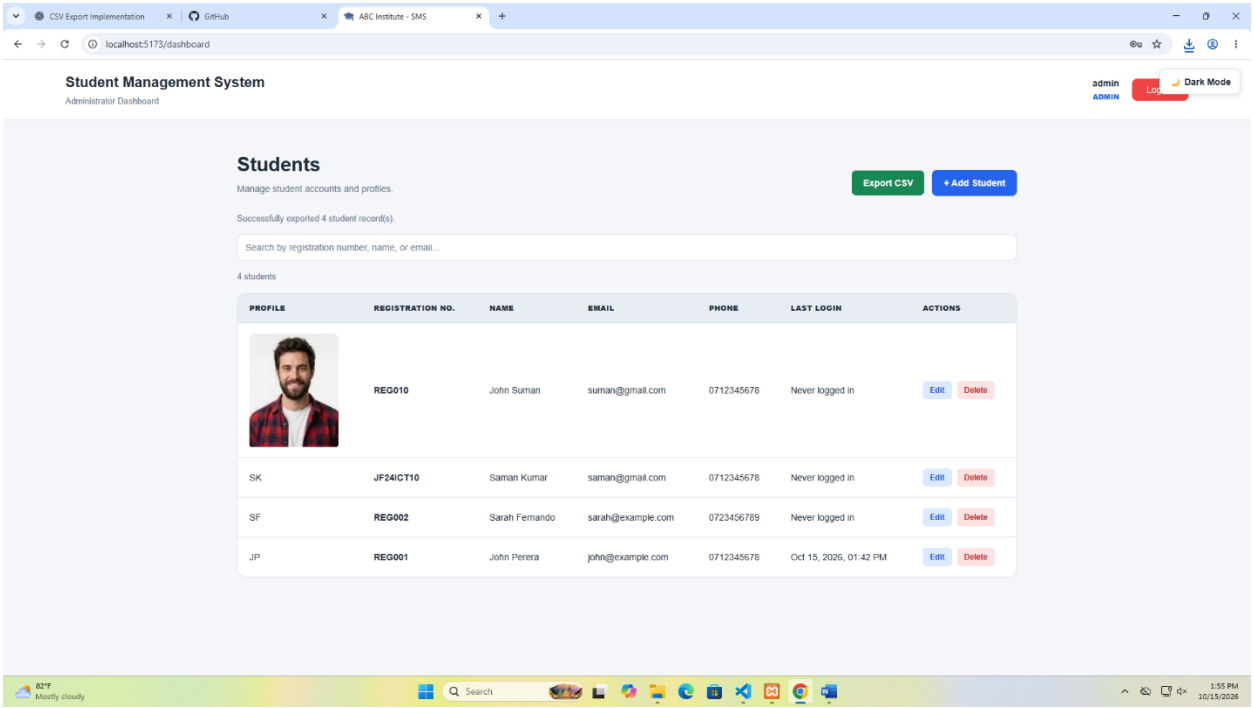


Figure 1: Administrator dashboard displaying the Export CSV button and the downloaded students.csv file opened in a spreadsheet application.

Evidence 02 – Export Validation and Regression Verification

Related: AC-005, AC-006, AC-007, AC-008

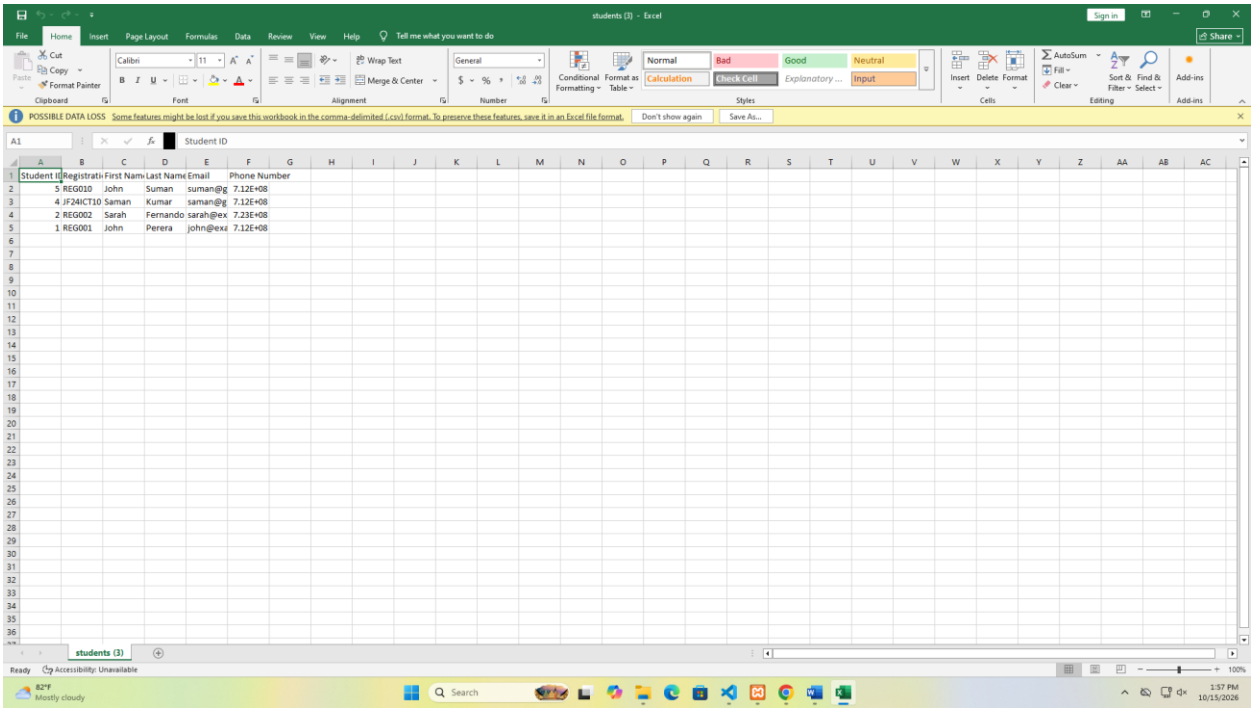


Figure 2: Evidence showing exported records with the search filter active, or CSV headers for an empty list, together with verification that existing student management functionality remains operational.

6. Final Result

Item	Result
Requirements	Pending Verification
Implementation	In Progress
Testing	Not Yet Completed
Acceptance Criteria	Pending Verification
Existing CRUD	Pending Verification
Final Status	Pending

6.1 Final Summary

CR-015 introduces a CSV export facility for the administrator's student dashboard. The planned implementation enables administrators to export the complete loaded student list into a downloadable students.csv file containing Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.

The implementation is designed to preserve valid CSV formatting, handle an empty student list, and retain existing student management functionality. The feature uses the existing administrator-protected student-list API and does not require a database schema change.

Final acceptance will be determined after implementation is completed and the defined test cases and acceptance criteria have been verified.

6.2 Issues / Limitations

Testing has not yet been completed. The final report must be updated with the actual implementation status, test results, acceptance-criteria verification, and any remaining issues after testing.

Student Name: J. Thushany

Student ID: JF24ICT09

Date: 15 October 2026

Signature: _____

~ END OF REPORT ~